

Four Models, One Stack: Training the Full Perception–Reasoning–Action Pipeline for Autonomous Drones

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The domain detector was the first model. It answered one question: can the drone see the things that matter — other drones, people from altitude, landing pads — instead of the 80 COCO categories it was born knowing? The answer was yes, and we documented the class-collapse problem it exposed.

After that we trained three more models in the same session. This post covers what they are, why we built each one, and how they all run together on a Jetson Orin Nano today.

Model 2 — VLM action-LoRA

The drone's reasoning layer uses a vision-language model to look at the camera feed, read the mission context, and emit a structured action: *move forward 3 m, photograph the target, land*. The stock Qwen3-VL-2B does this reasonably well but it was trained on the internet — not on drone missions.

We fine-tuned Qwen2.5-VL-3B-Instruct with LoRA (rank 16) on 2,000 drone-aerial training examples. Training ran for 3 epochs on hoopoe (2× RTX 5090) with 4-bit BitsAndBytes quantization. Final training loss: 0.31. Token accuracy on a held-out 200-example validation split: 94.4%.

The merged model was converted to GGUF Q4_K_M via llama.cpp: first a float16 GGUF, then quantized. Output: 1.8 GB main model + 1.2 GB multimodal projector. Both slot directly into the existing on-device VLM path.

Model 3 — Reactive policy MLP

The VLM makes good mission-level decisions but it runs at 1–3 Hz. At cruise speed a drone covers 1.5–3 m per inference cycle. That is fine for high-level waypoint selection but too slow for obstacle avoidance and altitude hold.

We trained a small MLP (two 128-unit hidden layers with LayerNorm, approximately 50 K parameters) to clone the planner. The input is a 12-dimensional state vector. The output is a 3-axis velocity command.

We generated 200,000 synthetic state-action pairs by sampling random states and running the planner rules as a function. 30 epochs of AdamW. Best validation MSE: 0.0077. The model fits in 121 KB as an ONNX file and can sustain 200 Hz on a Jetson Nano.

Model 4 — Monocular depth fine-tune

The quadcopter and rover use a stereo depth camera (Intel D435i) for ranging. Stereo baseline works well up to about 10–15 m. Beyond that, depth is unavailable.

We fine-tuned Depth Anything V2 Small on aerial imagery using self-distillation: run the zero-shot model on 500 VisDrone val images to generate pseudo-depth labels, then fine-tune on those labels.

10 epochs, scale-invariant SmoothL1 loss. The output is a 1.6 MB ONNX. Input: 518×518 RGB. Output: 518×518 relative depth map. Metric scaling is done at runtime using the known flight altitude as a prior.

Deployment

All four models are running on both the rover (Jetson Orin Nano) and the quadcopter. The deployment sequence: (1) Copy updated daemon.py and model setup scripts via SCP. (2) Run setup script to pull model files from S3. (3) Restart the drone-api service.

All models are publicly available on Hugging Face at [astralhf/astral-drone-models](https://huggingface.co/astralhf/astral-drone-models), including domain detector, VLM LoRA, reactive policy, and depth models.